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DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Project Poseidon: Intelligent Ocean Health Prediction Using Satellite Data and MLOps

Machine Learning Operations – AI254TA

TUTORIAL REPORT

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ABSTRACT

Oceans play a vital role in regulating Earth's climate, supporting biodiversity, and sustaining human life by absorbing atmospheric carbon dioxide, moderating global temperatures, and providing food and livelihoods to billions of people. However, rapid industrialization, increased greenhouse gas emissions, and unsustainable human activities have severely degraded ocean health. Critical issues such as ocean acidification, deoxygenation, coral bleaching, and harmful algal blooms are becoming more frequent and intense, posing serious threats to marine ecosystems and coastal communities.

Traditional ocean monitoring systems rely on in-situ physical sensors such as buoys, research vessels, and underwater instruments. Although these methods provide accurate localized measurements, they are expensive, geographically limited, and unable to deliver continuous global monitoring. The resulting data is often fragmented and delayed, limiting early detection and prediction of environmental risks.

Project **Poseidon – AI for Ocean Health Monitoring** proposes an intelligent, automated, and scalable system that leverages satellite-derived datasets and machine learning techniques to monitor, analyze, and predict ocean health conditions. The system integrates key ocean parameters such as sea surface temperature, chlorophyll concentration, pH, and dissolved oxygen to generate predictive insights and early warnings.

An end-to-end MLOps pipeline is implemented to ensure reproducibility, scalability, and reliability. Dataset versioning is handled using DVC, experiment tracking is performed using MLflow, and data drift detection is carried out using Evidently AI. The trained models are deployed as RESTful APIs using FastAPI, containerized using Docker, and deployed on the cloud using Render. By combining real-world datasets, machine learning, and modern MLOps practices, Project Poseidon provides a proactive solution for ocean health monitoring and supports the United Nations Sustainable Development Goals SDG 13 (Climate Action) and SDG 14 (Life Below Water).

1. INTRODUCTION

1.1 Overview of the Problem

Oceans are a fundamental component of Earth's life-support system, covering more than 70% of the planet's surface. They regulate climate patterns, absorb carbon dioxide, store heat, and support diverse ecosystems essential for food security, transportation, and

economic development. Despite their importance, oceans are experiencing rapid degradation due to climate change and human-induced environmental stressors.

One of the most serious challenges affecting ocean health is **ocean acidification**, caused by increased absorption of atmospheric carbon dioxide, which lowers seawater pH and disrupts marine ecosystems. Coral reefs and shell-forming organisms are particularly vulnerable, leading to cascading effects across marine food chains. Another major issue is **ocean deoxygenation**, where rising sea temperatures reduce oxygen solubility and nutrient pollution accelerates oxygen consumption, resulting in hypoxic “dead zones” that threaten marine life and fisheries.

Conventional monitoring systems rely on physical sensors such as buoys and research vessels. While accurate, these systems are costly, spatially limited, and often provide intermittent data. As a result, large portions of the global ocean remain poorly monitored. Satellite remote sensing offers global and continuous observation of ocean parameters such as sea surface temperature and chlorophyll concentration; however, satellite data is complex and requires advanced analytical techniques to extract meaningful insights. This creates the need for intelligent systems capable of integrating multi-source data, analyzing trends, and predicting future ocean conditions.

1.2 Motivation

The motivation for this project arises from the growing urgency to address climate change and its impact on marine ecosystems. Rising ocean temperatures, increasing acidity, and declining oxygen levels threaten coral reefs, fish populations, and coastal livelihoods worldwide. Existing monitoring approaches are largely reactive, identifying damage only after it has occurred, which limits opportunities for preventive action.

There is also a lack of effective decision-support systems that convert complex ocean data into actionable insights. Most existing platforms provide raw data or static visualizations that require expert interpretation. Advances in machine learning, cloud computing, and MLOps provide an opportunity to develop scalable, automated, and predictive ocean monitoring systems that can bridge this gap. Academically, this project enables practical application of AIML concepts to a real-world environmental problem.

1.3 Business Objective

The primary objective of Project Poseidon is to design and deploy a scalable, automated, and intelligent ocean health monitoring system that delivers reliable insights in real time. The system aims to continuously monitor key ocean parameters, predict future conditions, and provide early warnings for risks such as coral bleaching, hypoxia, and harmful algal blooms.

To support informed decision-making, the system introduces a simplified **Ocean Health Score (0–100)** that enables non-technical stakeholders to quickly assess ocean conditions. Automation and reliability are achieved through a comprehensive MLOps pipeline using tools such as Git, DVC, MLflow, Evidently AI, Docker, and FastAPI, with cloud-based deployment for global accessibility.

1.4 Scope of the Project

This project includes the use of real satellite-derived and sensor-based datasets, data preprocessing, exploratory data analysis, feature engineering, and training of multiple machine learning and deep learning models. It also incorporates experiment tracking, dataset versioning, drift detection, API-based backend development, containerization, and cloud deployment.

Future enhancements include real-time satellite API integration, advanced visualizations, voice-based AI assistance, multi-language support, and integration with government and international environmental platforms.

2. PROBLEM STATEMENT AND CHALLENGES

2.1 Problem Statement

The major problems addressed by Project Poseidon are listed below:

- The health of global oceans is declining rapidly due to climate change, pollution, and human activities.
- Critical ocean issues such as:
 - Ocean acidification
 - Ocean deoxygenation
 - Rising sea surface temperatures
 - Coral reef degradationare increasing in frequency and severity.
- Existing ocean monitoring systems are:
 - Expensive to deploy and maintain
 - Limited to specific geographic regions
 - Unable to provide continuous global monitoring

- Most systems depend on physical sensors such as buoys and research vessels, which:
 - Cover only small ocean areas
 - Generate delayed and fragmented data
- Large volumes of satellite ocean data are available, but:
 - Data is complex and high-dimensional
 - Manual analysis is inefficient
 - Predictive insights are rarely generated
- Current solutions lack:
 - Predictive capability
 - Automated monitoring
 - Explainable AI outputs
 - Robust MLOps pipelines

Formal Problem Definition

- To design and develop an **AI-powered, automated, and scalable ocean health monitoring system** that:
 - Integrates real satellite and sensor data
 - Predicts future ocean conditions
 - Provides explainable and actionable insights
 - Maintains reliability using MLOps practices
 - Is accessible through cloud-based deployment

2.2 Key Challenges Identified

- **Data Availability and Integration Challenges:**
 Ocean data collected from satellites and sensors often contains missing values, noise, and irregular sampling. Integrating multi-source datasets is complex due to differences in data formats, spatial resolution, and temporal resolution.
- **Data Quality and Variability Challenges:**
 Ocean datasets exhibit outliers, inconsistent units, and incomplete records.

Seasonal and regional variations introduce high variability, requiring extensive preprocessing before modeling.

- **Modeling Complexity Challenges:**

Ocean systems are highly nonlinear and influenced by multiple interacting factors. Traditional statistical models fail to capture long-term trends, while deep learning models require large datasets, high computational resources, and pose interpretability challenges.

- **Scalability and Automation Challenges:**

Models must process large volumes of continuously incoming satellite data. Manual retraining is infeasible, making automated pipelines for training, monitoring, and deployment essential.

- **Deployment and Maintenance Challenges:**

Machine learning systems often fail in production due to environment mismatches, dependency conflicts, and lack of monitoring, requiring containerization, cloud deployment, and API-based serving.

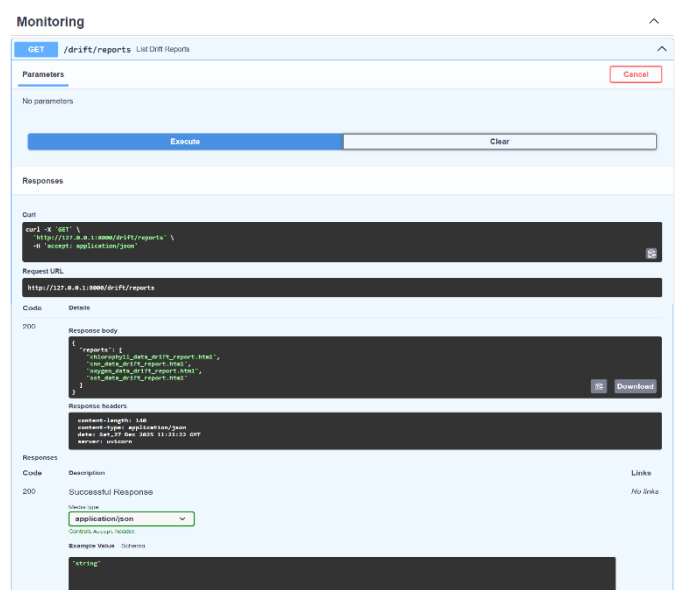
- **Trust and Explainability Challenges:**

Many AI models act as black boxes. Environmental decision-making requires transparency and scientific justification, and lack of explainability reduces trust and adoption by policymakers.

2.3 Proposed Solution Overview

Project Poseidon addresses the above challenges by:

- Using **real satellite-derived datasets** for global coverage.
- Applying **machine learning and deep learning models** for prediction.
- Implementing **EDA and feature engineering** for data quality.
- Using **MLOps tools** to ensure:
 - Reproducibility
 - Monitoring
 - Scalability
- Deploying the system using:
 - FastAPI for backend



- Docker for containerization
 - Render for cloud hosting
- Providing **explainable outputs** and simplified health metrics.

2.4 Benefits of the Proposed Approach

- Continuous and large-scale ocean monitoring
- Early warning for environmental risks
- Reduced dependency on physical sensors
- Automated and reproducible ML lifecycle
- Cloud-accessible and scalable system
- Decision-friendly outputs for non-technical users

3. DATASET COLLECTION

3.1 Overview of Dataset Collection

- Project Poseidon uses **real-world ocean datasets** collected from internationally recognized scientific agencies.
- The datasets are selected to study key ocean health indicators related to:
 - Ocean acidification
 - Ocean deoxygenation
 - Sea surface temperature rise
 - Chlorophyll concentration and algal blooms
- Unlike small academic projects, Poseidon integrates **multiple datasets from different domains**, making it a **multi-source and multi-modal system**.
- The datasets include:
 - Numerical tabular data
 - Time-series data
 - Satellite-derived imagery
- This approach ensures:

- Better generalization of models
- Higher scientific reliability
- Improved prediction accuracy

3.2 Objectives of Dataset Collection

- To obtain global-scale ocean observations rather than localized measurements.
- To ensure datasets are authentic, publicly available, and scientifically validated.
- To support machine learning, time-series forecasting, and image-based deep learning.
- To enable long-term trend analysis, drift detection, and reproducible experimentation.

3.3 Categories of Datasets Used

The datasets used in Project Poseidon are categorized as follows:

3.3.1 Physical Ocean Datasets

- Sea Surface Temperature (SST)
- Ocean heat stress indicators

3.3.2 Chemical Ocean Datasets

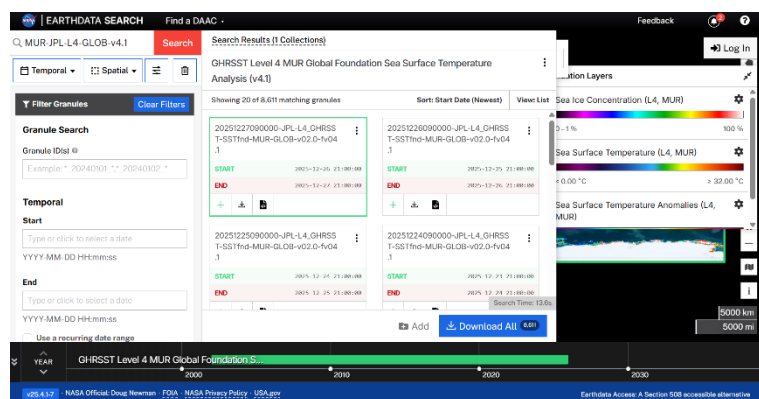
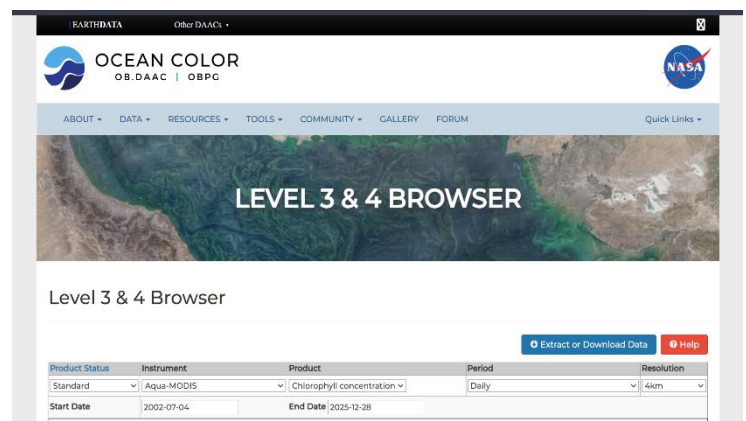
- Dissolved oxygen concentration
- Ocean pH levels
- CO₂-related chemical indicators

3.3.3 Biological Ocean Datasets

- Chlorophyll-a concentration
- Algal bloom indicators

3.3.4 Satellite Image Datasets

- Ocean color images
- Chlorophyll heat maps
- Sea surface temperature maps



3.4 Dataset Sources, Usage, and Management

- Datasets were sourced from NASA, NOAA, and GOA-ON, using satellite missions such as MODIS and sensor platforms like ARGO floats.
- Tabular datasets were stored in CSV format for machine learning and time-series models, while satellite images were processed for CNN-based analysis.
- Data was used for:
 - LSTM-based forecasting (SST and oxygen)
 - Random Forest and XGBoost models (pH and chlorophyll)
 - CNN-based image analysis
- All datasets were organized into data/raw and data/processed directories and versioned using DVC and Git, ensuring reproducibility and auditability.

3.5 Summary of Datasets Used

Dataset	Source	Format	Model Used
SST	NASA / NOAA	CSV	LSTM
Oxygen	NOAA	CSV	LSTM
pH	NOAA / GOA-ON	CSV	Random Forest
Chlorophyll (Tabular)	NASA	CSV	RF / XGBoost
Satellite Images	NASA	PNG	CNN

3.6 Key Strengths of Dataset Design

- Uses **multiple real-world data sources**
- Combines:
 - Satellite data
 - Sensor data
- Supports:
 - Tabular ML
 - Time-series forecasting

- Deep learning
- Fully reproducible using DVC and Git

4. EXPLORATORY DATA ANALYSIS (EDA)

4.1 Tools and Technologies Used for EDA

The following tools were used during the EDA phase:

- **Pandas**
 - Data loading and manipulation
 - Handling missing values
 - Summary statistics
- **NumPy**
 - Numerical operations
 - Array-based computations
- **Matplotlib**
 - Line plots
 - Histograms
 - Trend visualization
- **Seaborn**
 - Correlation heatmaps
 - Scatter plots
- **Jupyter Notebook**
 - Interactive analysis
 - Visual validation
- **DVC**
 - Versioning raw and processed datasets
- **MLflow**
 - Logging EDA-related metadata and parameters

4.2 Data Quality Assessment and Preprocessing

- Missing values were identified using standard null-value checks.

- Causes of missing data included:
 - Cloud cover in satellite imagery
 - Temporary sensor outages
 - Data transmission gaps
- Numerical time-series data was handled using:
 - Linear interpolation
 - Forward and backward filling
- Corrupted satellite images were excluded to maintain data integrity.

4.3 Outlier Detection and Data Consistency Checks

- Outliers were detected using:
 - Box plots
 - Z-score analysis
- Outliers were primarily caused by:
 - Sensor malfunction
 - Extreme environmental events
- Mild outliers were retained to preserve real-world variability.
- Extreme outliers were clipped using percentile thresholds.
- Data consistency checks ensured:
 - Valid ranges
 - Unit consistency
 - Removal of duplicate and invalid records

Drift is detected for 0.0% of columns (0 out of 8).

Column	Type	Reference Distribution	Current Distribution	Data Drift	Stat Test	Drift Score
> label	cat			Not Detected	Z-test p_value	1
> mean_g	num			Not Detected	K-S p_value	1
> std_g	num			Not Detected	K-S p_value	1
> range	cat			Not Detected	chi-square p_value	0.999863
> std_b	num			Not Detected	K-S p_value	0.998954
> std_r	num			Not Detected	K-S p_value	0.98794
> range	cat			Not Detected	chi-square p_value	0.999863
> std_b	num			Not Detected	K-S p_value	0.998954
> std_r	num			Not Detected	K-S p_value	0.98794
> mean_b	num			Not Detected	K-S p_value	0.975762
> mean_r	num			Not Detected	K-S p_value	0.956898

Rows per page: 8 rows
 < 1 of 8 >

- Spatial analysis showed:
 - Higher chlorophyll near coastlines
 - Warmer SST near equatorial regions
- Correlation analysis highlighted:
 - Strong negative correlation between SST and oxygen
 - Moderate positive correlation between SST and chlorophyll
 - Negative relationship between pH and CO₂ indicators
- Distribution analysis showed skewness in chlorophyll data, corrected using log transformations.
- EDA confirmed:
 - Ocean temperature as a dominant driver
 - Interdependence among ocean parameters
 - Suitability of processed data for ML and DL modeling

4.5 Dataset-Wise Exploratory Analysis

Dataset	Key Observations	Insights Derived
Sea Surface Temperature (SST)	Values ranged between 18°C–34°C with strong seasonal patterns	Rising SST linked to oxygen depletion and coral bleaching
Dissolved Oxygen	Declining trends in warmer regions with seasonal fluctuations	Strong negative correlation with SST
Ocean pH	Gradual downward trend observed	Indicates ongoing ocean acidification
Chlorophyll (Tabular)	High values near coastlines and seasonal spikes	Early indicator of algal blooms and nutrient pollution
Satellite Images	Noise, blurring, and missing regions observed	Required normalization and resizing (224×224) for CNN models

5. FEATURE ENGINEERING

5.1 Types of Features Engineered

To improve model performance and capture complex ocean dynamics, the following feature categories were engineered:

- Temporal features
- Statistical features
- Environmental interaction features
- Spatial features
- Image-based features
- Reduced-dimensional features

5.2 Temporal and Statistical Feature Engineering

- Temporal features were extracted from timestamps, including:
 - Year
 - Month
 - Day of year
 - Seasonal indicators
- These features capture strong seasonal behavior in ocean data.
- Statistical features were generated using rolling windows:
 - Rolling mean
 - Rolling standard deviation
 - Rolling minimum and maximum
- These features help smooth noisy signals and capture short-term fluctuations and long-term trends, especially for time-series models such as LSTM.

5.3 Environmental Interaction Features

- Interaction features were created by combining multiple ocean parameters to represent environmental stress.
- Examples include:

- Temperature–oxygen ratio
 - Acidification index (CO_2 / pH)
 - Heat stress indicators
- These features capture nonlinear relationships between ocean variables and significantly improve model expressiveness.

5.4 Spatial Feature Engineering

- Geographic coordinates (latitude and longitude) were used directly as features.
- Additional spatial transformations included:
 - Region-based encoding
 - Distance from coastline
- Spatial features improved regional prediction accuracy, particularly for coastal zones with higher pollution and chlorophyll concentration.

5.5 Feature Scaling, Encoding, and Selection

- Feature scaling was applied using:
 - StandardScaler (zero mean, unit variance)
 - Min–Max scaling for bounded features
- Categorical features such as seasons were one-hot encoded.
- Feature selection was performed using:
 - Random Forest feature importance
 - Correlation thresholds
- Redundant and low-impact features were removed to balance accuracy and interpretability.

5.6 Image-Based Feature Engineering (CNN)

- Satellite images were preprocessed using:
 - Resizing to 224×224 pixels
 - Pixel normalization

- Noise reduction
- Convolutional Neural Networks automatically extracted:
 - Spatial patterns
 - Texture and color variations
- These learned features represent chlorophyll density and algal bloom patterns effectively.

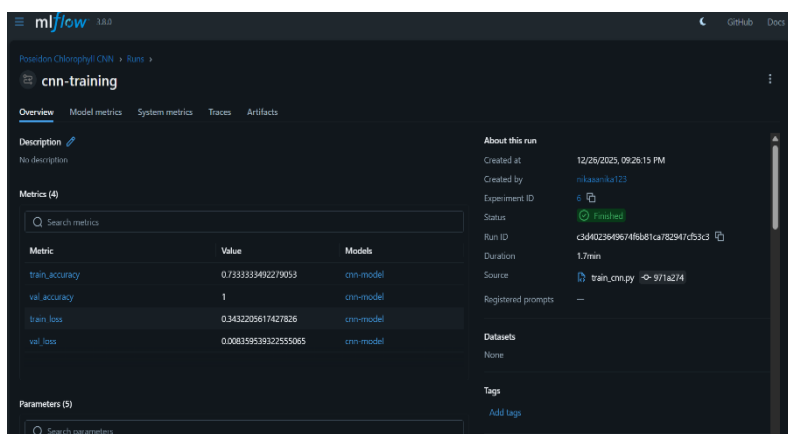
5.7 Dimensionality Reduction (PCA)

- Principal Component Analysis (PCA) was applied to:
 - Reduce feature redundancy
 - Improve computational efficiency
- PCA was mainly used for:
 - Drift detection
 - Feature visualization
 - Simplifying model inputs

6. MACHINE LEARNING MODELS AND TRAINING

6.1 Multi-Model Approach

- Project Poseidon adopts a multi-model strategy to handle heterogeneous ocean data.
- Reason:
 - Tabular data, time-series data, and satellite images exhibit different patterns.
- Both machine learning and deep learning models are used to ensure robustness and adaptability.

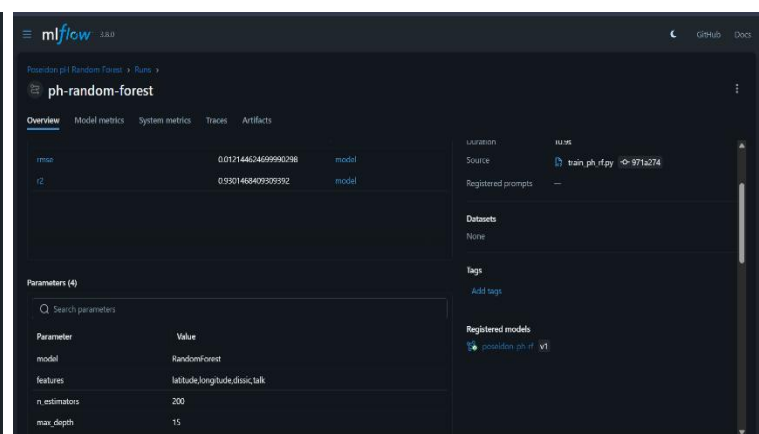


The screenshot shows the mlflow UI for a run named 'cnn-training'. The interface includes tabs for Overview, Model metrics, System metrics, Traces, and Artifacts. The Overview tab is active, displaying a table of metrics and a sidebar with run details.

Metric	Value	Models
train_accuracy	0.733333402279053	cnn-model
val_accuracy	1	cnn-model
train_loss	0.343205617427826	cnn-model
val_loss	0.00835953932255065	cnn-model

About this run

- Created at: 12/26/2023, 09:26:15 PM
- Created by: mlbasanku123
- Experiment ID: 6
- Status: finished
- Run ID: c3d4023649674f8b61ca782941cf53c3
- Duration: 1.7min
- Source: train_cnn.py @ 971a27f4
- Registered prompts: —
- Datasets: None
- Tags: Add tags

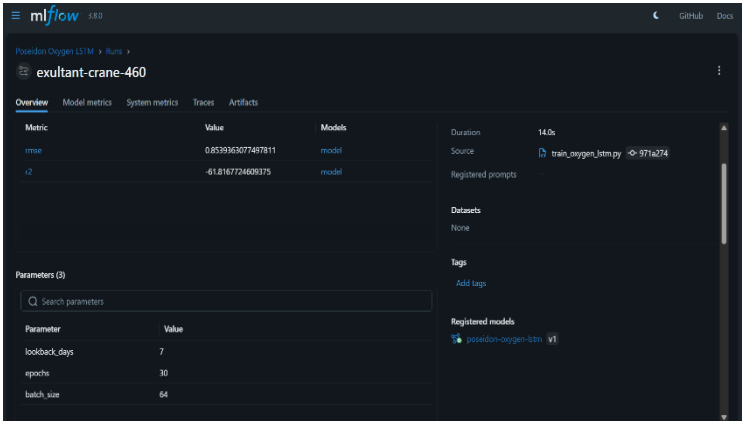
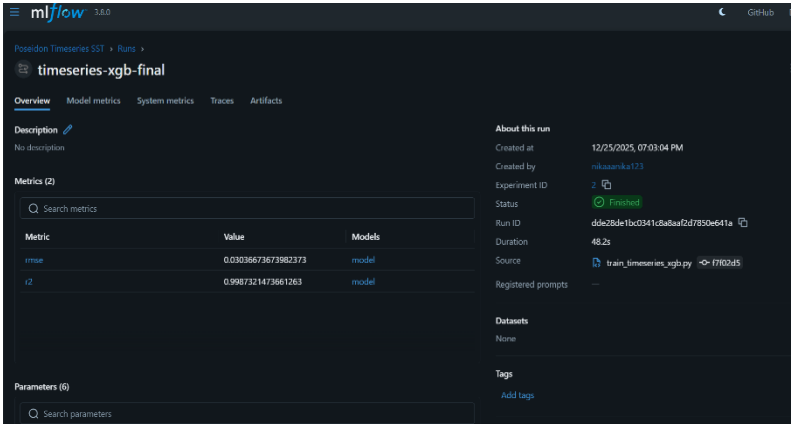


The screenshot shows the mlflow UI for a run named 'ph-random-forest'. The interface includes tabs for Overview, Model metrics, System metrics, Traces, and Artifacts. The Overview tab is active, displaying a table of metrics and a sidebar with run details.

Parameter	Value
model	RandomForest
features	latitude,longitude,disic,talk
n_estimators	200
max_depth	15

About this run

- Created at: 0.01214462469990298
- Created by: 03901468409305932
- Experiment ID: 6
- Status: finished
- Run ID: c3d4023649674f8b61ca782941cf53c3
- Duration: 1.7min
- Source: train_ph_rf.py @ 971a27f4
- Registered prompts: —
- Datasets: None
- Tags: Add tags
- Registered models: poseidon-ph-rf v1



6.2 Models Used and Their Purpose

Model	Data Type	Primary Purpose
Random Forest	Tabular	Ocean Health Score prediction
XGBoost	Tabular	Chlorophyll prediction and bloom risk
LSTM	Time-series	Forecasting SST and dissolved oxygen
CNN	Image data	Algal bloom and spatial anomaly detection

6.3 Random Forest and XGBoost Models

- **Random Forest**
 - Used for Ocean Health Score computation (0–100).
 - Handles nonlinear relationships and noisy environmental data.
 - Provides feature importance for explainability.
- **XGBoost**
 - Used for chlorophyll concentration prediction.
 - Gradient boosting improves accuracy and generalization.
 - Regularization minimizes overfitting.
- Both models were trained on feature-engineered tabular datasets.

6.4 Deep Learning Models (CNN and LSTM)

- **CNN**
 - Used for satellite image analysis.
 - Extracts spatial patterns, texture, and color variations.
 - Effective in detecting chlorophyll-rich regions and harmful algal blooms.
- **LSTM**
 - Used for time-series forecasting of SST and dissolved oxygen.
 - Captures long-term dependencies and seasonal trends.
 - Suitable for early warning of deoxygenation and temperature rise.

6.5 Model Training and Hyperparameter Optimization

- Models were trained using:
 - Train–test split
 - Time-aware splitting for LSTM
- Hyperparameters optimized included:
 - Tree depth and estimators (RF, XGBoost)
 - Learning rate
 - Number of layers and batch size (CNN, LSTM)
- Training was performed in isolated environments to ensure consistency.

6.6 Model Evaluation Strategy

- Model evaluation was performed to assess:
 - Prediction accuracy
 - Generalization capability
 - Stability under varying data conditions
- Different evaluation strategies were used based on data type and model architecture.

6.6.1 Evaluation Techniques

- Train-Test Split
 - Used for initial performance validation.
- Cross-Validation
 - Applied to tabular models (Random Forest, XGBoost) to reduce overfitting.
- Time-Aware Validation
 - Used for LSTM models to preserve temporal order and prevent data leakage.
- Hold-out Image Validation
 - Used for CNN models to evaluate unseen satellite images.

6.6.2 Evaluation Metrics Used

Model Type	Metrics Used	Purpose
Regression Models (RF, XGBoost, LSTM)	RMSE, MAE, R^2	Measure prediction accuracy and error
Classification Models (CNN)	Accuracy, Precision, Recall	Evaluate bloom detection reliability
Time-Series Models	RMSE, Trend Accuracy	Assess forecast stability

6.6.3 Model Performance Insights

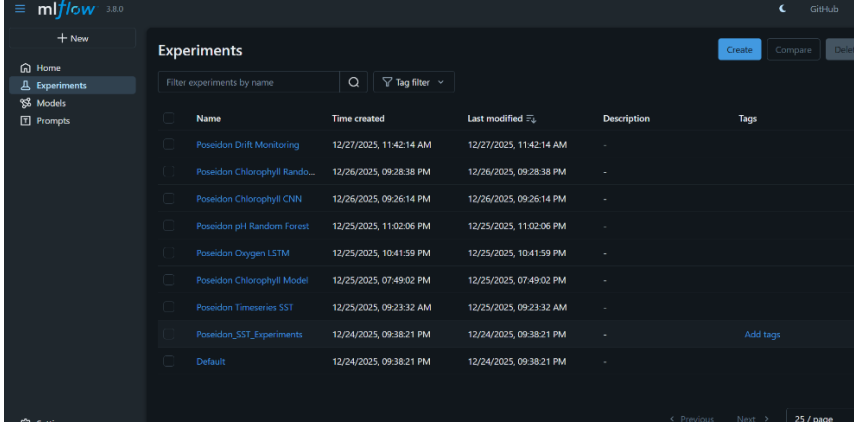
- Random Forest provided:
 - Stable predictions
 - High interpretability
- XGBoost achieved:
 - Lower RMSE
 - Better generalization on unseen data
- LSTM successfully:
 - Captured seasonal and long-term trends

```
(poseidon-env) nikaanika123@nikasssst:~/poseidon-elops$ mlflow ui
Backend store URI not provided. Using sqlite:///mlflow.db
Registry store URI not provided. Using backend store URI.
2025/12/28 09:24:44 INFO mlflow.store.db.utils: Creating initial MLflow database tables...
2025/12/28 09:24:44 INFO mlflow.store.db.utils: Updating database tables
2025/12/28 09:24:44 INFO alembic.runtime.migration: Context impl SQLiteImpl.
2025/12/28 09:24:44 INFO alembic.runtime.migration: Will assume non-transactional DDL.
2025/12/28 09:24:44 INFO alembic.runtime.migration: Context impl SQLiteImpl.
2025/12/28 09:24:44 INFO alembic.runtime.migration: Will assume non-transactional DDL.
2025/12/28 09:24:44 INFO mlflow.store.db.utils: Creating initial MLflow database tables...
2025/12/28 09:24:44 INFO mlflow.store.db.utils: Updating database tables
2025/12/28 09:24:44 INFO alembic.runtime.migration: Context impl SQLiteImpl.
2025/12/28 09:24:44 INFO alembic.runtime.migration: Will assume non-transactional DDL.
[WARNING] Security middleware enabled with default settings (localhost-only). To allow connections from other hosts, use
--host 0.0.0.0 and configure --allowed-hosts and --cors-allowed-origins.
INFO: uvicorn running on http://127.0.0.1:5000 (Press CTRL+C to quit)
INFO: Started parent process [2036]
INFO: Started server process [2038]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: Started server process [2039]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: Started server process [2041]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: Started server process [2039]
INFO: Waiting for application startup.
INFO: Application startup complete.
```

- **CNN effectively:**
 - **Detected spatial anomalies**
 - **Identified chlorophyll-rich regions**

6.6.4 Role of MLflow in Evaluation

- **MLflow was used to:**
 - **Log evaluation metrics**
 - **Compare multiple model runs**
 - **Select best-performing models**
- **Ensured:**
 - **Transparent comparison**
 - **Reproducible evaluation results**



The screenshot shows the MLflow Experiments page. On the left is a sidebar with navigation links: Home, Experiments (selected), Models, and Prompts. The main area is titled 'Experiments' and contains a search bar and a 'Tag filter' dropdown. Below this is a table listing various experiments. Each row includes a checkbox, the experiment name, the time it was created, the last time it was modified, a description, and a 'Tags' column. The experiments listed include 'Poseidon Drift Monitoring', 'Poseidon Chlorophyll Rando...', 'Poseidon Chlorophyll CNN', 'Poseidon pH Random Forest', 'Poseidon Oxygen LSTM', 'Poseidon Chlorophyll Model', 'Poseidon Timeseries SST', 'Poseidon_SST_Experiments', and a 'Default' row. At the bottom right of the table, there is a link that says 'Add tags'.

	Name	Time created	Last modified	Description	Tags
☐	Poseidon Drift Monitoring	12/27/2025, 11:42:14 AM	12/27/2025, 11:42:14 AM	-	
☐	Poseidon Chlorophyll Rando...	12/26/2025, 09:28:38 PM	12/26/2025, 09:28:38 PM	-	
☐	Poseidon Chlorophyll CNN	12/26/2025, 09:26:14 PM	12/26/2025, 09:26:14 PM	-	
☐	Poseidon pH Random Forest	12/25/2025, 11:02:06 PM	12/25/2025, 11:02:06 PM	-	
☐	Poseidon Oxygen LSTM	12/25/2025, 10:41:59 PM	12/25/2025, 10:41:59 PM	-	
☐	Poseidon Chlorophyll Model	12/25/2025, 07:49:02 PM	12/25/2025, 07:49:02 PM	-	
☐	Poseidon Timeseries SST	12/25/2025, 09:23:32 AM	12/25/2025, 09:23:32 AM	-	
☐	Poseidon_SST_Experiments	12/24/2025, 09:38:21 PM	12/24/2025, 09:38:21 PM	-	Add tags
☐	Default	12/24/2025, 09:38:21 PM	12/24/2025, 09:38:21 PM	-	

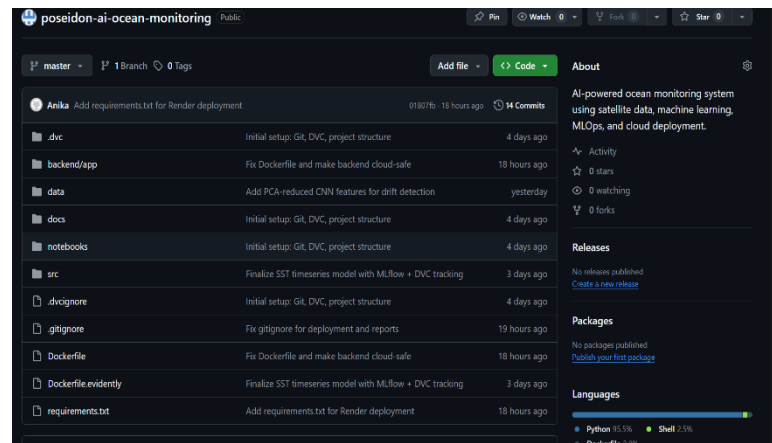
6.6.5 Importance of Model Evaluation

- **Prevents deployment of weak or biased models.**
- **Ensures:**
 - **Scientific validity**
 - **Reliable real-world performance**
- **Forms the basis for:**
 - **Model selection**
 - **Drift monitoring reference data**

6.7 Model Selection and Versioning

- MLflow was used to compare experiments and select best-performing models.
- Selection criteria included:

- Accuracy
- Stability
- Interpretability
- Computational cost
- Final models were versioned using:
 - MLflow (artifacts and metadata)
 - DVC (large model files)
- Enables reproducibility and safe rollback.



7. MLOps PIPELINE AND TOOLS

7.1 End-to-End MLOps Pipeline

- Project Poseidon implements a fully automated end-to-end MLOps pipeline to manage the complete ML lifecycle.
- The pipeline covers:
 - Data ingestion
 - Dataset versioning
 - Model training and evaluation
 - Experiment tracking
 - Deployment
 - Monitoring and drift detection
- Automation ensures scalability, reproducibility, and long-term maintainability in a continuously changing ocean environment.

7.2 Source Code Version Control (Git)

- Git is used to version control:
 - ML training scripts
 - Feature engineering code

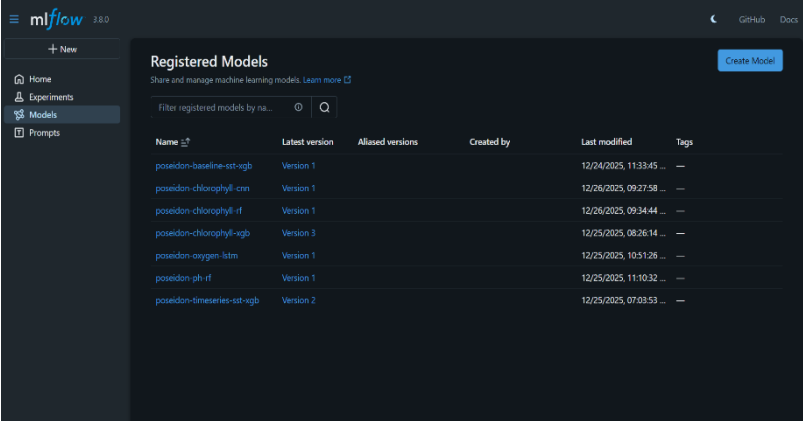
- Backend APIs
- Docker and deployment configurations
- Maintains complete change history, enables rollback to stable versions, and supports collaboration.
- GitHub integration enables CI/CD-based cloud deployment.

7.3 Dataset Versioning Using DVC

- DVC is used to manage large and frequently updated datasets.
- Tracks versions of:
 - Raw datasets
 - Processed and feature-engineered datasets
 - Large model files
- Stores only metadata in Git, keeping the repository lightweight.
- Enables exact dataset reproduction and experiment rollback using git checkout and dvc checkout.

7.4 Experiment Tracking and Model Management (MLflow)

- MLflow is used to track:
 - Model types
 - Hyperparameters
 - Evaluation metrics
 - Dataset version references
- Enables visual comparison of experiments and selection of best-performing models.
- Ensures scientific reproducibility and transparent experimentation.

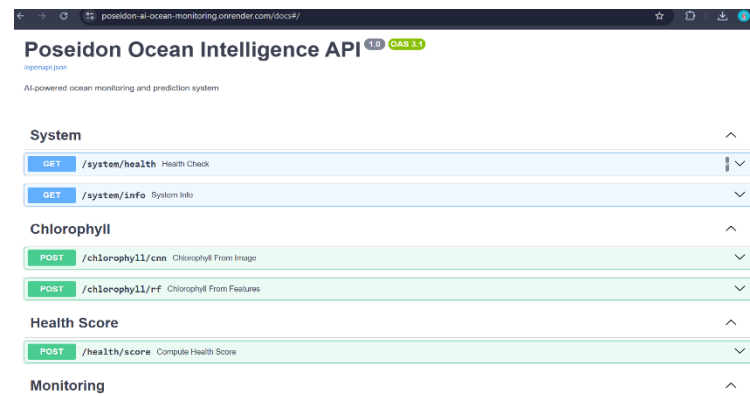


Name	Latest version	Aliased versions	Created by	Last modified	Tags
poseidon-baseline-stt-rgb	Version 1			12/24/2025, 11:33:45 ...	—
poseidon-chlorophyll-con	Version 1			12/26/2025, 09:27:58 ...	—
poseidon-chlorophyll-rl	Version 1			12/26/2025, 09:34:44 ...	—
poseidon-chlorophyll-rgb	Version 3			12/25/2025, 08:26:14 ...	—
poseidon-oxygen-lstm	Version 1			12/25/2025, 10:51:26 ...	—
poseidon-ph-rl	Version 1			12/25/2025, 11:10:32 ...	—
poseidon-timeseries-stt-rgb	Version 2			12/25/2025, 07:03:53 ...	—

7.5 Model Serving and Deployment (FastAPI + Docker + Render)

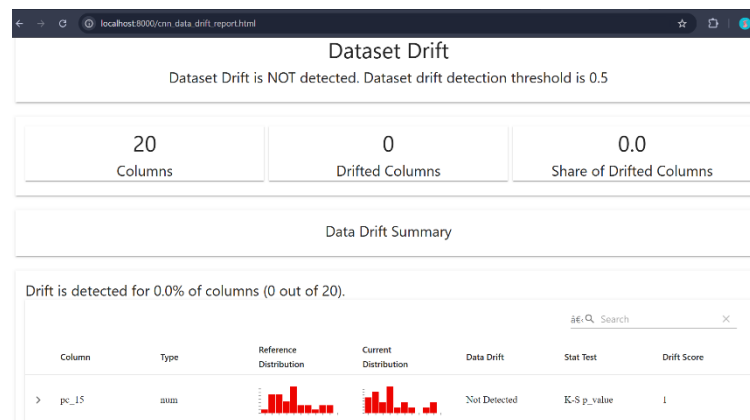
- Trained models are exposed as REST APIs using FastAPI.
- FastAPI provides:
 - High-performance inference

- Built-in request validation
 - Automatic API documentation
- Docker is used to containerize the backend, ensuring environment consistency and portability.
- Render is used for cloud deployment due to:
 - Docker support
 - GitHub integration
 - Automatic redeployment on code changes



7.6 Monitoring and Drift Detection (Evidently AI)

- Continuous monitoring is implemented to handle changing ocean data patterns.
- Evidently AI is used to detect:
 - Data drift
 - Feature drift
 - Prediction drift
- Generates interactive HTML reports for model health assessment.
- Drift detection enables early warnings and supports retraining decisions.



7.7 Automation, Security, and Reliability

- The combined use of Git + DVC + MLflow + Docker ensures:
 - Automated experiments
 - Reproducible results
 - Consistent deployment
- APIs are protected using authentication mechanisms and role-based access.
- Logging and error handling improve system reliability and production readiness.

8. DEPLOYMENT, MONITORING, AND RISK MANAGEMENT

8.1 Deployment Architecture

- Project Poseidon follows a cloud-based deployment architecture to ensure global accessibility.
- Trained models are served as RESTful APIs using FastAPI.
- The backend handles:
 - Model loading
 - Inference requests
 - Drift report serving

8.2 Containerization and Cloud Deployment

- Docker is used to containerize the complete backend, including:
 - ML models
 - Dependencies
 - Runtime environment
- Containerization ensures:
 - Environment consistency
 - Easy rollback
 - Platform independence
- Render is used for cloud deployment due to:
 - Native Docker support
 - GitHub integration
 - Automatic redeployment on code updates

8.3 Monitoring and Drift Detection

- Continuous monitoring is essential due to changing ocean conditions.
- Evidently AI is used to detect:
 - Data drift

- Feature drift
 - Prediction drift
- Interactive HTML reports provide early warning of model degradation and support retraining decisions.

8.4 Data and Model Risks

- Data Risks
 - Missing or noisy data due to sensor failures and satellite interference
 - Mitigated using preprocessing, interpolation, and multi-source data integration
- Model Risks
 - Overfitting and concept drift
 - Mitigated using cross-validation, MLflow tracking, and drift monitoring

8.5 Deployment and Infrastructure Risks

- Risks include:
 - Dependency conflicts
 - Resource limitations on free cloud tiers
- Mitigation strategies:
 - Docker-based isolation
 - Lightweight models
 - Pre-deployment testing

8.6 Security and Reliability Measures

- APIs secured using authentication and role-based access.
- Logging and error handling implemented for fault detection.
- Versioned models enable safe rollback and stable long-term operation.

9. CONCLUSION AND FUTURE WORK

Project Poseidon demonstrates the effective application of Artificial Intelligence and MLOps practices for large-scale ocean health monitoring. By integrating real-world satellite and sensor datasets, the system successfully analyzes and predicts key ocean parameters such as sea surface temperature, dissolved oxygen, pH levels, and chlorophyll concentration. The use of multiple machine learning and deep learning models enables the system to handle heterogeneous data types and capture complex environmental patterns. The implementation of a complete MLOps pipeline ensures reproducibility, scalability, reliable deployment, and continuous monitoring, making the system suitable for real-world environmental applications.

Overall, the project provides a proactive and data-driven approach to understanding ocean health and identifying early warning signs of environmental stress. By transforming complex scientific data into interpretable metrics and predictive insights, Project Poseidon supports informed decision-making for researchers, policymakers, and environmental organizations. The system contributes directly to global climate initiatives, particularly the United Nations Sustainable Development Goals SDG 13 (Climate Action) and SDG 14 (Life Below Water), and establishes a strong foundation for future advancements such as real-time monitoring, advanced visualizations, and large-scale deployment.